

Work Experience

Software Development Engineer

Jan 2025 – Present

Amazon Web Services Transcribe | Seattle, WA

Java, C++, AWS, Ruby, Triton Inference Server, Cuda, Docker

- Improved GPU Automatic Speech Recognition (ASR) model inference throughput by ~4× using Triton Inference Server and optimizing shared Cuda memory, increasing GPU utilization by ~15% under peak traffic across concurrent streams.
- Built routing to serve GPU models with request validation, allowlisting, periodic canary tests, alarms, and automatic CPU fallback to maintain availability.
- Integrated Custom Vocabulary support for GPU inference by generating locale specific embeddings for customer defined terms.
- Shipped Language Identification model with expanded locale coverage by fine tuning weights to reduce misclassifications.
- Implemented continuous cache and parallel decoding in the CPU ASR model, reducing word error rate by ~10%.
- Supported ARR growth from ~\$50M to ~\$100M by increasing peak traffic capacity by ~40% through multiple instance types and reducing cold start latency by ~20% through predictive scaling and warm pools.
- Benchmarked word error rate across three ASR model variants using spoken, written, punctuation, custom vocabulary, and noisy datasets to identify statistically significant regressions prior to launch.

Software Development Engineer

Sep 2024 – Dec 2024

Amazon Leo | Redmond, WA

Python, PyTest, Shell Scripting, Grafana, AWS (S3, Athena), Docker

- Built a production testing framework to command and configure all satellite venues, enabling coordinated end-to-end validation across modulation frequency schemes.
- Enabled simulation of satellite contact scenarios by triggering GPS spoofing, increasing coverage without requiring live satellite passes.
- Improved data processing performance by ~72% by creating PySpark tables to summarize CQM and MAC data, automating legacy ETL workflows.

Undergraduate Research Assistant

Aug 2023 – May 2024

Purdue Computational Quantum Electromagnetics Lab | West Lafayette, IN

Python, NumPy, SciPy, Matplotlib

- Designed algorithms to generate Hamiltonian matrices for configurable quantum systems with multiple transmon qubits and cavities.
- Solved Schrödinger's equation using ODE based numerical methods to analyze quantum system dynamics.

Software Development Engineer Intern

May 2022 – Aug 2022; May 2023 – Aug 2023

Amazon Leo | Redmond, WA

Python, SQL, PyTest, Grafana, Docker

- Improved satellite subsystem validation efficiency by ~70% using statistical and time-series telemetry models across hardware and software components.
- Built real-time anomaly detection and forecasting into production test automation for early issue detection.
- Developed dashboards using antenna and cloud telemetry to simulate orbital modem passes and surface corrective actions.
- Provided network traffic insights to leadership, documented model architecture, and trained end users for adoption.

Skills

Programming: Java, Python, C/C++, Ruby, TypeScript, SQL, Shell Scripting, HTML, CSS, Javascript

ML Infrastructure: Triton Inference Server, TensorFlow, Cuda

Infrastructure: AWS (EC2, SageMaker, ElastiCache, CloudFormation), Docker, CI/CD, Spark, PyTest, Grafana, Git

Education

Purdue University

West Lafayette, IN

B.S. Computer Science

Dec 2024

Certifications

AWS Cloud Practitioner Essentials, Amazon ML University MLOps, Amazon ML University Time Series Forecasting

Projects

Bitcoin Reinforcement Learning Trading Bot

Aug 2020 – May 2021

Python, TensorFlow, Cuda, NumPy, Pandas | [GitHub](#)

- Built a Cuda-accelerated Deep Recurrent Q-Network trading bot achieving 74% ROI on historical Bitcoin data.